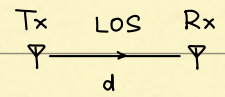
Path Loss 路径损耗

$$P_L = \frac{P_t}{P_r}, \quad P_L[dB] = 10 \lg P_L = P_t[dB] - P_r[dB]$$

Free-Space Path Loss 自由空间路径损耗

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2 \propto d^{-2}$$



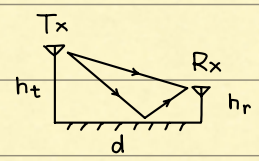
$$P_L[dB] = -10 \log G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$

三种传播机制：

reflection 反射， diffraction 衍射， scattering 散射

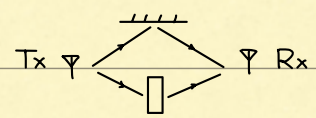
2. Two-Ray Racing

$$P_r = P_t G_t G_r \left(\frac{h_t h_r}{d^2}\right)^2 \propto d^{-4}$$



Multipath 多径效应

3. Simplified Path Loss Model



$$P_r = P_t K \left(\frac{d_r}{d}\right)^{\gamma} \propto d^{-\gamma}$$

d_r : reference distance $\quad d_r = 1\,m$

γ : loss exponent $\quad \gamma \in [2, 8]$

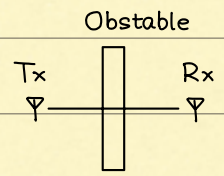
K : path loss

$$P_r[dB] = P_t[dB] + K[dB] - 10\gamma \log \frac{d}{d_r}$$

$$P_L[dB] = -10\gamma \log \frac{d_r}{d} - K[dB]$$

Shadowing Fading 阴影衰落

路径损耗确定性，阴影衰落为随机变量



$$\psi[dB] = P_t[dB] - P_r[dB] \sim N(\mu_\psi[dB],\ \sigma^2)$$

$$\psi = \frac{P_t}{P_r} \sim \text{对数正态}$$

同时考虑 Path Loss + Shadowing Fading

$$P_r = P_t K \left(\frac{d_r}{d}\right)^\gamma \cdot \psi$$

$$P_r[dB] = P_t[dB] + K[dB] + \psi[dB] - 10\gamma \lg \frac{d}{d_r}$$

$$\frac{P_t}{P_r}[dB] = P_L[dB] + \psi[dB]$$

$$= \underbrace{K[dB] + \psi[dB]}_{\sim N(K[dB],\ \sigma^2)} - 10\gamma \lg \frac{d}{d_r}$$

Outage Probability 中断概率

$$P_{out} = P(P_r < P_{min})$$

[例]

距离	P_t / P_r [dB]
5 m	-60 dB
25 m	-80 dB

(1) MMSE 估计 γ，$K[dB]$

(2) 仅考虑 Path Loss，

$$P_r[dB] = P_{min} + 10\ dB$$

求中断概率 P_{out}

(1) $\min\ (\mu_{\tilde{K}} - 10\gamma \lg 5 + 60)^2$

$+ (\mu_{\tilde{K}} - 10\gamma \lg 25 + 80)^2$

解得 γ，$\mu_{\tilde{K}}$

(2) 代入得 $\psi[dB]$ 的估计 σ^2

$$P_{out} = P(\psi[dB] < -10\ dB)$$

衰落分类

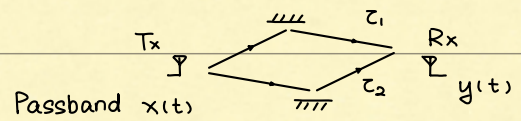
Multipath	Flat	$\sigma \ll T_s$	σ Delay Spread
		$W_c \gg W$	$W_c = \frac{1}{\sigma}$ Coherence Bandwidth
	Selective	$\sigma > T_s$	$D = \frac{v}{\lambda}$ Doppler Shift
		$W_c < W$	$T_c = \frac{0.423}{D}$ Coherence Time
Time-Vary	Slow	$D \ll W$	T_s Symbol Duration
		$T_c > T_s$	W Baseband
	Fast	$D \gg W$	
		$T_c < T_s$	

Multipath 多径效应

无线信道为时变系统，不同路径不同时延 τ_i

$\sigma = \tau_1 - \tau_2$: Delay Spread 时延扩展

类似于回音时长



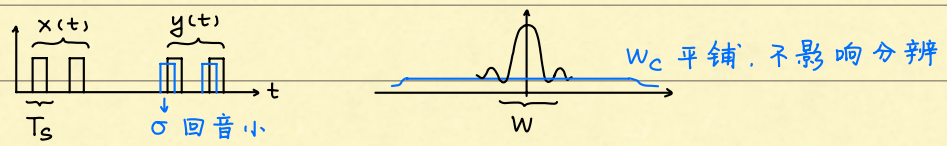
$W_c := \frac{1}{\sigma}$: Coherence Bandwidth 相干带宽

$x(t) \longrightarrow \boxed{h(t,\tau)} \longrightarrow y(t)$

Flat fading 平坦衰落

$\sigma \ll T_s$ (Symbol Duration), $W_c > W$ (Baseband)

不同路径差不多，不影响 0/1 之间分辨

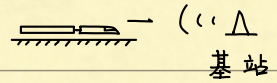


Selective fading 选择性衰落

ISI (Intersymbol Interference) 码间干扰，即回声

$\sigma > T_s$, $W_c < W$

Doppler Shift $D = \frac{v}{\lambda}$

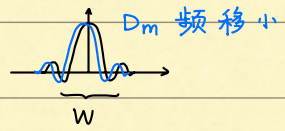


Coherence Time 相干时间 $T_c := \frac{0.423}{D}$

Slow Fading 慢衰落

$D \ll W$, $T_c \gg T_s$

几乎认为用户不移动，信道频移可忽略



Fast Fading 快衰落

$D > W$, $T_c < T_s$